

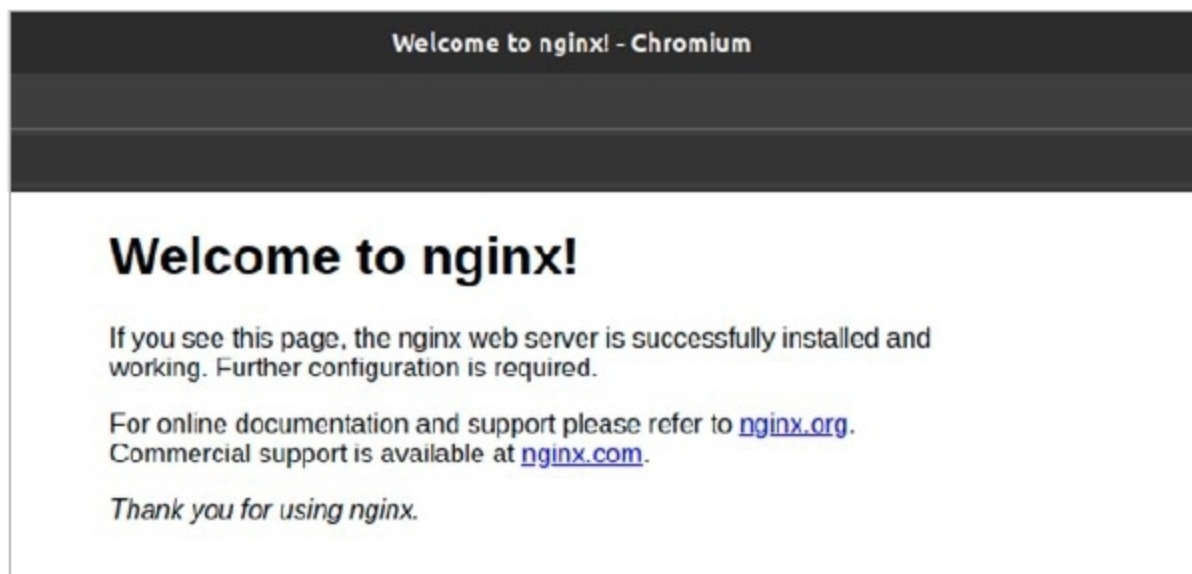


Figure 1: Nginx welcome page

- You have server access and you are connected to the server using `ssh`.

Section 1- Installing the Nginx Web server

The Nginx Web server is used to display Web pages to visitors. It is used as a reverse proxy, load balancer, mail proxy and HTTP cache.

In a Debian based OS, you can install any package using a simple `apt` command. The best practice is that whenever you deploy a server, you do an update and an upgrade with the following commands:

```
sudo apt-get update
sudo apt-get upgrade
```

Install the first package, Nginx, using the following:

```
sudo apt-get install nginx
```

You can check whether Nginx is installed properly or not by visiting the following URL:

```
http://public_ip
```

If you see the 'Welcome to Nginx' page, then the installation is done.

Note: You need to allow HTTP and HTTPS requests from your firewall if it's enabled.

Section 2 - Installing the MySQL server

MySQL is an open source relational database management system. Its name is a combination of 'My', the name of co-founder Michael Widenius's daughter, and 'SQL', the abbreviation for Structured Query Language.

To install the MySQL database, you need to execute the following command:

```
sudo apt-get install mysql-server
```

Configure MySQL. To secure the installation, execute one script. This script comes with the MySQL package.

Note: This `mysql_secure_installation` is optional.

If you enable the 'Validate password' plugin and if your password doesn't match the specified criteria, it will be rejected by MySQL with an error.

```
sudo mysql_secure_installation
```

```
Would you like to setup VALIDATE
PASSWORD plugin?
```

```
Press y|Y for Yes, any other key for
No: Y
```

```
There are three levels of password
validation policy:
```

```
LOW    Length >= 8
MEDIUM Length >= 8, numeric, mixed
case, and special characters
STRONG Length >= 8, numeric, mixed
case, special characters and dictionary
file
```

```
Please enter 0 = LOW, 1 = MEDIUM and 2
= STRONG: 1
```

```
Please set the password for root here.
```

```
New password:
```

```
Re-enter new password:
```

```
Do you wish to continue with the
password provided?(Press y|Y for Yes,
any other key for No) : y
```

```
Remove anonymous users? (Press y|Y for
Yes, any other key for No) : Y
```

```
Disallow root login remotely? (Press
y|Y for Yes, any other key for No) : Y
```

```
Remove test database and access to it?
(Press y|Y for Yes, any other key for
No) : Y
```

```
Reload privilege tables now? (Press y|Y
for Yes, any other key for No) : Y
```

```
All done!
```

For an Ubuntu system running MySQL 5.7 (or later versions), the root MySQL user is set to authenticate using the `auth_socket` plugin by default rather than with a password. Switch its authentication method from `auth_socket` to `mysql_native_password`. To do this, execute the following commands:

```
sudo mysql
```

```
mysql > SELECT user,authentication_
string,plugin,host FROM mysql.user;
```

```
mysql > ALTER USER 'root'@'localhost'
IDENTIFIED WITH mysql_native_password
BY 'password';
```

```
mysql > FLUSH PRIVILEGES;
```

```
mysql > SELECT user,authentication_
string,plugin,host FROM mysql.user;
```

```
mysql > exit
```